

2026 Jan 27

Two-dimensional bifurcation

Hopf bifurcation:

Ex. $\dot{x}_1 = x_1 (\mu - x_1^2 - x_2^2) - x_2$
 $\dot{x}_2 = x_2 (\mu - x_1^2 - x_2^2) + x_1$

In polar coordinates

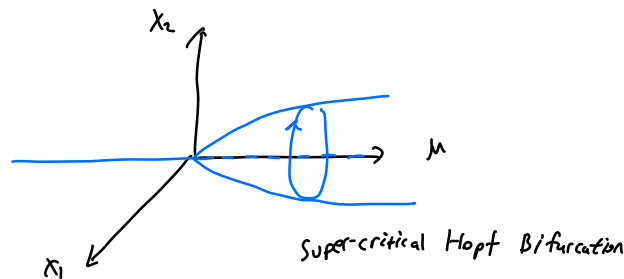
$$\begin{array}{l|l} \dot{r} = \underbrace{\mu r - r^3}_{f(r)} & f(r) = 0 \\ \dot{\theta} = 1 & r(\mu - r^2) = 0 \\ & r = 0 \text{ or } \mu - r^2 = 0 \end{array}$$

$\mu < 0$: equilibrium at $x = 0$ Stable

$\mu > 0$: equilibrium at $x = 0$ unstable

and periodic orbit at $r = \sqrt{\mu}$ stable
 (limit cycle)

$$\begin{array}{l} \dot{r} = \underbrace{\mu r - r^3}_{f(r)} \\ \frac{df}{dr} = \mu - 3r^2 \\ \frac{df}{dr} \Big|_{r=\sqrt{\mu}} = \mu - 3\mu = -2\mu < 0 \text{ for } \mu > 0 \end{array}$$



Ex. $\dot{r} = \underbrace{\mu r + r^3 - r^5}_{f(r)} \quad \Bigg| \quad r = 0 \text{ is always a solution to } f(r) = 0$
 $\dot{\theta} = 1$

Always an equilibrium at $x = 0$

$$f(r) = r(\mu + r^2 - r^4)$$

$$r^2 = z$$

$$z \geq 0$$

$$\mu + z - z^2 = 0$$

$$z = \frac{-1 \pm \sqrt{1+4\mu}}{2\mu}$$

$$\circ \mu > 0, \quad z = \frac{-1 + \sqrt{1+4\mu}}{2\mu} \quad r = \sqrt{z}$$

$$\circ -\frac{1}{4} < \mu < 0, \quad z = \frac{-1 \pm \sqrt{1+4\mu}}{2\mu} \quad r = \sqrt{z}$$

$$\circ \mu < -\frac{1}{4} \quad \text{no solution in } z \text{ and hence } r$$

$\mu > 0$: equilibrium of $x=0$

unstable

and a periodic orbit at $r = \sqrt{z_1}$

stable

$-\frac{1}{4} < \mu < 0$: equilibrium of $x=0$

stable

and two periodic orbit at $r = \sqrt{z_2} \quad z_2 < z_3$

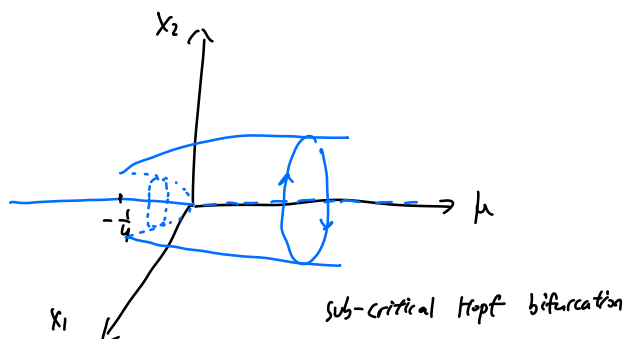
unstable

$$r = \sqrt{z_3}$$

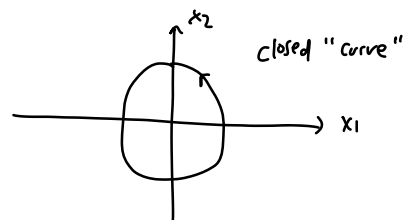
stable

$\mu < -\frac{1}{4}$: equilibrium

stable



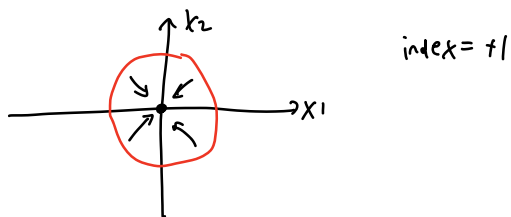
Index Theory (planar systems)



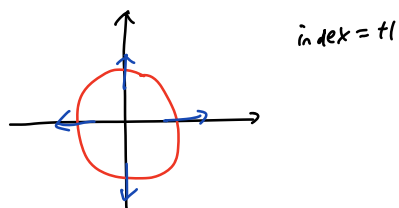
Index: The index of a closed curve is k if,
when traversing the curve in one direction, $f(x)$ rotates by $2\pi k$ in the same direction

For any closed orbit, $\text{index} = +1$

node (stable/unstable)



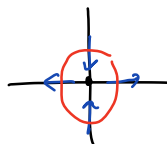
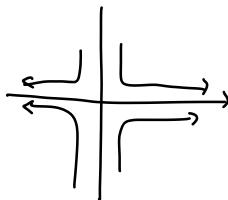
Unstable node



focus, center

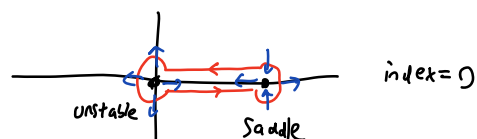
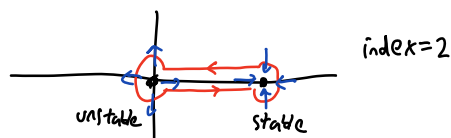
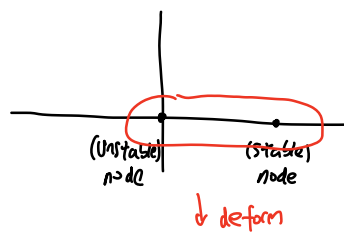
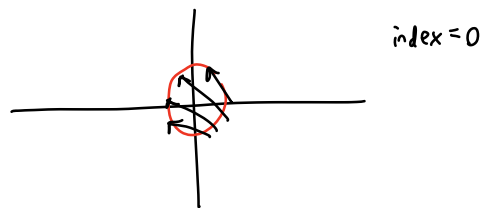
$\text{index} = +1$

Saddle-point



$\text{index} = -1$

No equilibrium inside the closed curve



Thm The index of a closed curve is equal to the sum of indices of the equilibria inside

Corollary Inside any periodic orbits, there must be at least one equilibrium

and indices of the equilibrium must add up to +1.

Recall example $\dot{x}_1 = x_2 = f_1(x_1, x_2)$

$\dot{x}_2 = -\delta x_2 + x_1 - x_1^3 + x_1^2 x_2 = f_2(x_1, x_2) \quad \delta > 0$

At equilibrium $x_2 = 0$

$$0 + x_1 - x_1^3 = 0 \rightarrow x_1 = 0, x_1 = \pm 1$$

Three equilibria: $(0, 0)$ $(1, 0)$ $(-1, 0)$

$$\frac{\partial f}{\partial x} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 - 3x_1^2 + 2x_1x_2 & -\delta + x_1^2 \end{bmatrix}$$

$$\frac{\partial f}{\partial x} \Big|_{(0,0)} = \begin{bmatrix} 0 & 1 \\ 1 & -\delta \end{bmatrix}$$

$$\begin{vmatrix} 1 & -\delta \end{vmatrix}$$

$\Downarrow$

e-values are real and have opposite signs

$\Rightarrow$ saddle

$$\frac{\partial f}{\partial x} \Big|_{\substack{(1,0) \\ (-1,0)}} = \begin{bmatrix} 0 & 1 \\ -2 & 1-\delta \end{bmatrix}$$

$\Downarrow$

e-values are real and the same sign

or are complex

$\rightarrow$ node, focus or center

